

Activity = 3

Analytical Problem:

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① Given $G_x = 24$
 $G_y = 18$

$$\begin{aligned}\text{Gradient magnitude} &= \sqrt{G_x^2 + G_y^2} \\ &= \sqrt{(24)^2 + (18)^2} \\ &= \sqrt{576 + 324} \\ &= \sqrt{900} \\ &= 30\end{aligned}$$

② Gradient direction: $\theta = \tan^{-1}(G_y/G_x)$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$\theta = \tan^{-1}(0.75)$$

$$\theta = 36.87^\circ$$

③ Given:

Pixel magnitude.

P_1 25

P_2 50

P_3 90

P_4 60

P_5 30

Solution:

$P_1 \rightarrow$ Border Pixel $\rightarrow$ suppressed (0)

$P_2 \rightarrow 50 \rightarrow$ Compare with 23 & 90
Not maximum so (0)

$P_3 \rightarrow 90 \rightarrow$ Compare with 80 & 60

It is maximum so (90)

$P_4 \rightarrow 60 \rightarrow$ Compare with 90 & 36

Not maximum so (0)

$P_5 \rightarrow 80 \rightarrow$ border so (0) suppressed

Pixel	P_1	P_2	P_3	P_4	P_5
Output	0	0	90	0	0

Strong edges: P_1, P_4

Weak edges: P_2, P_5

Discard edges: P_3, P_6

③ Given

High Threshold = 100

Low Threshold = 30

Pixel 2 magnitude

Classification

135

P_1

$$P_1 = 135 > 100$$

Strong edge

90

P_2

$$P_2 = 90 < 100$$

Weak

49

P_3

$$P_3 = 49 < 100$$

Discard

120

P_4

$$P_4 = 120 > 100$$

Strong

70

P_5

$$P_5 = 70 < 100$$

Weak

20

P_6

$$20 < 100$$

Discard.